

Incorporation*

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1 Introduction

This chapter examines the syntax of *noun incorporation*, a grammatical phenomenon whereby a nominal argument of a verb, typically its object, may be morphologically integrated into the verb complex. To start, consider the Mapudungun (Araucanian) examples below. In (1a), the nominal object *waka* ‘cow’ stands as an independent word, whereas in (1b) it is incorporated into the verb and the additional nominal material seen in (a) is absent. This alternation is moreover productive, with few lexical or semantic restrictions on the choice of verb or noun in question.

(1) *Canonical noun incorporation (Mapudungun)*

- a. Ñi chao kintu-le-y ta chi pu **waka**
my father seek-PROG-IND.3S.S the COLL cow
‘My father is looking for the cows.’
- b. Ñi chao kintu-**waka**-le-y
my father seek-cow-PROG-IND.3S.S
‘My father is looking for the cows.’ (Baker et al. 2005, p. 139, citing Salas 1992)

This chapter overviews typological and theoretical investigations of the grammatical profile illustrated above, which I refer to in this paper as ‘canonical’ noun incorporation, following Mithun 1984, 1986 and others. Additionally, it considers incorporation to include ‘pseudo’ noun incorporation, in which the noun-verb relationship is based on linear adjacency rather than affixation (Masam, 2001; Dayal, 2011, a.o.), as well as ‘obligatory’ noun incorporation, which arises when the

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incorporation process is required, rather than optional, for certain verbs or certain nouns (Sadock, 1980; Johns, 2007; Wojdak, 2008; Yuan, 2025, a.o.). Within each of these profiles, we find additional points of cross-linguistic variation. It is therefore an open question whether incorporation is a unified phenomenon to begin with, or whether it is simply a useful descriptive label for a family of surface-similar patterns. Nonetheless, syntactic research on incorporation has generally been guided by a stable set of theoretical questions, crosscutting the abovementioned distinctions. One such question concerns the division of labor at the syntax-morphology interface: to what extent is incorporation driven by the abstract sentence-level principles regulating the distribution of nominal arguments? This is, however, complicated by a lack of consensus pertaining to the grammatical mechanism(s) responsible for incorporation (and for complex word-formation, more broadly), as well as to whether incorporation itself is a syntactic phenomenon to begin with (as opposed to a postsyntactic, phonological, or lexical one).

Throughout this chapter, I also highlight various phenomena that are overall less integrated in the literature on this topic, yet still highly relevant to the issues introduced above. These include the doubling of elements syntactically (or at least semantically) associated with the incorporant (e.g. Rosen, 1989; Haugen, 2009; Barrie, 2015) and the availability of incorporants for syntactic processes such as ϕ -agreement, case assignment, and movement (e.g. Baker et al., 2005; Farkas and de Swart, 2003; Barrie and Mathieu, 2016; van Urk, 2020; Yuan, 2025). Moreover, although the majority of the literature on incorporation has focused on that of nominal objects, incorporation may sometimes instead apply to nominals of other grammatical functions, as well as to non-nominal elements such as modifiers, verbs, and even clauses (e.g. Pittman, 2006, 2009; Wojdak, 2008; Barrie and Li, 2012; Yuan, 2026). Existing analyses of incorporation attempt to accommodate (all or a subset of) these patterns in different ways, and it is not immediately obvious how generalizable any given account may be beyond its immediate empirical domain.

2 Canonical noun incorporation

2.1 Morphosyntactic overview

Wilhelm von Humboldt first introduced the concept of incorporation (*Einverleibung*), in reference to the Nahuatl (Uto-Aztecan) construction *ni-naka-kwa* ‘I eat meat’ (von Humboldt, 1836); like in Mapudungun, (1), this instantiates a canonical incorporation pattern in that it alternates with a semantically similar construction with a standalone object. For von Humboldt, incorporation was a purely morphological property (see also Sapir 1911 for a similar early perspective). The idea that incorporation is not syntactic is also replicated in more modern lexicalist accounts, which hold that incorporation constructions are created in the lexicon as verb-noun compounds (e.g. Sciullo and Williams, 1987; Rosen, 1989; Spencer, 1995; Anderson, 2000). In contrast, and as detailed further below, the modern generative tradition largely takes incorporated nominals to be created in the syntax as thematic arguments of the verbs, into which they are ultimately morphologically integrated (see also Kroeber 1911). Under this view, incorporation constructions are *derived* and are thus distinct from compounds. That being said, this lexical–syntactic debate effectively disappears within frameworks such as Distributed Morphology (Halle and Marantz, 1993), which take *all* types of word-building (even compounding; Harley 2008) to be syntactic in nature; see Massam 2009b for further discussion of this point as it relates to incorporation.

In a seminal paper on incorporation, Mithun 1984 offers a four-way typology that organizes subtypes of incorporation along an implicational hierarchy, meaning that a given language that displays a type higher along the hierarchy also displays the lower types. Note that this typology excludes the obligatory incorporation phenomena discussed later in this chapter, which Mithun does not take to be incorporation to begin with. In Type I, which Mithun refers to as compounding, the incorporated noun is non-referential, the resulting verb is morphosyntactically intransitive, and the process is not necessarily productive. In Type II, incorporation results in an apparent manipulation of case or argument status, for instance by promoting a dative (non-core) argument to absolutive (core) upon incorporating the direct object, as shown in (2) with Alutor (Chukotko-

Kamchatkan).

(2) *Type II noun incorporation and case manipulation (Alutor)*

- a. gəm-nan akka-ŋ tə-nə-svitku-və-tk-ən **utte-ʔut**
I-ERG son-DAT 1SG.A-CAUS-cut-SUFF-PRES-3SG.P wood-ABS
'I am making the son cut wood.'
- b. gəm-nan akək tə-n-**u**-svitku-və-tk-ən
I-ERG son.ABS 1SG.A-CAUS-wood-cut-SUFF-PRES-3SG.P
'I am making the son cut wood.' (Podobryaev 2013, p. 287, citing Koptjevskaja-Tamm and Muravyova 1993)

Type III uses incorporation in discourse to background non-salient nominals, and, unlike in Types I-II, the incorporated noun may be interpreted as specific or definite. According to Baker et al. 2005, Mapudungun falls under this type, given textual examples like (3). Finally, in Type IV the incorporated noun functions as a classifier and may co-occur with an independent object (discussed below).

(3) *Type III noun incorporation as argument backgrounding (Mapudungun)*

- Kiñe kelluwen rëtre-ke-Ø-y **pali** ñi tripalwi pële kangelu ingkawen
one team push-3O-IND.3SS ball 3.POSS goal toward other side
katrütu-**pali**-ke-y
intercept-ball-HAB-3SS
'One team pushes the ball toward their goal, and the other side tries to intercept it.' (Baker et al. 2005, p. 145, citing Harmelink 1992)

Other generalizations have been made about the properties of incorporated nominals. For instance, it is often said that incorporation involves *detransitivization* (e.g. Mardirussian, 1975; Baker, 1988), which may be seen through changes relating to case, agreement, and transitivity morphology. In Koryak (Chukotko-Kamchatkan), an ergative-absolutive language, incorporation of the direct object co-occurs with the absence of both object-indexing ϕ -agreement and ERG case morphology on the (logical) transitive subject, (4). In other languages, incorporation may also result in a change in transitivity-encoding verbal morphology.

(4) *Case and agreement changes in incorporation (Koryak)*

- a. jejyut̪cew̪əɭʔ-ə-jək na-ko-jəɭŋ-ə-ŋ-na-w **kali-w**
 student-EP-OBL.NSG.ERG INV-PRES-read-EP-PRES-3.O-3PL book-ABS.PL
 ‘The students are reading books.’
- b. jejyut̪cew̪əɭʔ-u Ø-ko-**kale**-jəɭŋ-al-la-ŋ-Ø
 student-ABS.PL 2/3.S/A.IND-PRES-book-read-VBLZ-PL-PRES-3.S.IND
 ‘The students are reading books.’ (Abramovitz 2021, p. 112)

Under many syntactic accounts of incorporation, the above properties are explained by appealing to the *structural size* of the incorporant. Indeed, a prevailing idea within this body of work holds that these nominals are syntactically smaller than standalone ones (e.g. Heck and Richards, 2010; Roberts, 2010; Barrie and Mathieu, 2016; Branigan and Wharram, 2019); for instance, if a nominal argument of a given language is normally a DP, then its incorporated counterpart must minimally lack this outermost DP layer (whether it is instead identified as a N(P) or some other category like *nP* depends on the individual analysis). This is independently motivated by the observation that incorporated nouns often cannot be proper names, pronouns, or possessive noun phrases—all indicators of DP structure. The morphosyntactic changes shown above follow if the relevant processes (e.g., case assignment and ϕ -agreement) specifically target DPs and ignore smaller nominals (e.g., Chomsky 2000, 2001). Moreover, if semantic information associated with referentiality is located in the DP layer, its absence in incorporated contexts accounts for the indefinite and low-scope interpretations typical of such nouns (e.g., van Geenhoven 1998; Chung and Ladusaw 2004, 2021).

However, this line of analysis is insufficient in capturing the full range of incorporation phenomena. It does not readily account for languages that fall under Mithun’s (1984) Type III, as Type III incorporated nouns may be semantically equivalent to their non-incorporated counterparts (Baker et al., 2005; Baker, 2009). Baker (1996) additionally observes that morphological detransitivization does not occur in all Type III and IV languages. For this reason, not all analyses of incorporation assume a size distinction to begin with; see, for instance, Haugen 2009 and den Dikken and Dékány 2021 for analyses involving incorporated DPs. And others such as Barrie and Mathieu (2016) propose that this may be a point of cross-linguistic parameterization, with differ-

ent languages permitting incorporants with varying sizes. For instance, (5) suggests that Meskwaki (Algonquian) allows DP-incorporation, since the incorporated argument in this sentence is a possessive DP (note also that the verb still bears transitive morphology, glossed ‘VTA’). The incorporation of full DPs has also been argued to be attested in languages with pseudo and obligatory incorporation (Levin, 2015; Yuan, 2025) (see §§3.2, 4.1).

(5) *Incorporation of possessive DP (Meskwaki)*

ne-pyätci-**ke-tānes-a**-wâpag-ā-pen

1-come.to-your-daughter-NMLZ-see-VTA-IMP.PL

‘We have come to see your daughter.’ (Barrie and Mathieu 2016, p. 19, citing Michelson 1917)

Another complication arises with languages that permit stranded articles and demonstratives, such as Onondaga (Iroquoian), (6). Although such elements could plausibly be analyzed as D⁰s (e.g. Abney, 1987), they could also be specifiers or adjuncts within NPs (e.g. Baker, 1988)—an analytical distinction that has consequences for one’s proposed theory of noun incorporation, as discussed below (see, for instance, (7)).

(6) *Incorporation may strand demonstrative (Onondaga)*

wa[?]-k-**nakt**-a-hnin[?]-[?] **neke[?]**

FACT-1SG.NOM-bed-EPEN-buy-PUNC this

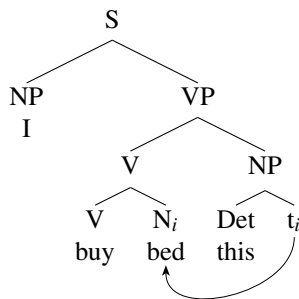
‘I bought this bed.’ (Barrie, 2009, p. 3)

Finally, as Salzmann (2020) points out, the DP vs. NP distinction in the context of (non-)incorporation is only meaningful insofar as the standalone arguments in the language are DPs to begin with (see Bošković 2005, 2008, Pereltsvaig 2007, Bruening 2020, a.o., for varying views surrounding this debate)—so it is not necessarily a given that the morphosyntactic and semantic properties of incorporants could be attributable to a difference in structural size or category alone.

2.2 A survey of analyses of canonical incorporation

Many—though not all—analyses within the generative framework take incorporation to involve *syntactic movement* in some capacity. The first major account in this vein comes from [Baker \(1988\)](#), who argues that incorporation is derived by syntactic head movement in the sense of [Travis 1984](#), as schematized in (7) (based on data like (6)). This analysis assumes, and in turn provides evidence for, the Uniformity of Theta Assignment Hypothesis (UTAH), which states that thematic roles are associated with dedicated syntactic positions ([Chomsky 1981](#); [Baker 1997](#)). Putting these together, this explains why the noun undergoing incorporation is most often a direct object or theme: as the complement of a verb, only a theme may undergo N-to-V head movement. In addition, this account predicts that, for languages that permit noun incorporation into intransitive verbs, such verbs are necessarily unaccusative rather than unergative. Baker demonstrates that this is indeed the case in languages such as Kanien’kéha [Mohawk] (Iroquoian).

(7) *Deriving incorporation via head movement*



Furthermore, this account captures modifier stranding: as (7) shows, these elements are simply left within the NP complement of V. However, as alluded to above, stranded determiner-like elements such as demonstratives cannot be analyzed as Ds that head their own DP projections, as such heads would block N-to-V head movement (the Head Movement Constraint; [Travis 1984](#)). Thus, [Baker \(1996\)](#) instead treats these elements as adjuncts within NP.

Baker’s account, however, may not readily explain why syntactically higher nominals *may* incorporate in some languages. Applied arguments encoding paths, locations, and instruments may also productively undergo incorporation (e.g. [Mithun, 1984](#); [Barrie and Li, 2012](#); [McKenzie, 2022](#)), though these elements could also be taken to be introduced low in the verbal domain (e.g.

Pylkkänen, 2008). Perhaps more problematic is the existence of agent incorporation, uncommon but attested in languages such as Sora (Munda) (Phillips, 2016; Anderson, 2017).¹

Rosen (1989) additionally points out that Baker’s head movement analysis cannot explain why some languages do not permit modifier stranding at all—a problem, given that modifier stranding has been used as evidence against lexicalist compounding approaches (cf. **a [happy dog]house*). Rosen argues instead that incorporating constructions are created in the lexicon, but there is cross-linguistic variation in how exactly incorporation affects the verb’s argument structure. In some languages, incorporation functions to semantically restrict the true object of the verb, which is unincorporated. Thus, apparent modifier stranding is simply *classifier incorporation*. Under this treatment, given examples such as (6), the determiner would be reanalyzed as the object, with the incorporated noun as its classifier (i.e., ‘I bed-bought this (one)’). See also §2.3 below for discussion of hyponymous doubling, which Rosen takes to also result from classifier incorporation.

In response, Baker et al. (2005) argue that a head movement analysis may in fact capture Rosen’s observation, if the nature of the copy (or trace) left by head movement may vary across languages. They posit a syntactic parameter governing the deletion of movement copies, by either preserving or removing their ϕ -features. If these features are preserved after head movement, they may license modifier stranding, as well as permit ϕ -agreement with incorporated objects, as in Southern Tiwa (Tanoan), (8) (the aforementioned Iroquoian languages are also analyzable in this vein).² In contrast, if the ϕ -features are deleted after head movement, then both modifier stranding and object ϕ -agreement become impossible. Note, however, that incorporation in Southern Tiwa is obligatory (rather than canonical) in certain contexts; see §4.2.

(8) *Object agreement with incorporated noun (Southern Tiwa)*

- | | |
|--|--|
| <p>a. Ti-seuan-mu-ban
 1 SG:A-man-see-PST
 ‘I saw the/a man.’</p> | <p>b. Bi-seuan-mu-ban
 1 SG:B-man-see-PST
 ‘I saw men.’ (Baker et al. 2005, p. 152)</p> |
|--|--|

¹Further examples of agent incorporation are given in §3.3 for languages with pseudo incorporation.

²Levin (2018) further extends this line of analysis, proposing that further morphosyntactic distinctions seen across incorporating languages could be captured in terms of genuine ϕ -agreement (feature valuation) vs. pronominal clitic-doubling with the incorporated object. See also §4.1 for further discussion of ϕ -agreement with incorporated nominals (Johns 2009; Yuan 2025).

That said, the validity of any head movement approach is complicated by the uncertain status of head movement in modern syntactic theory. Given several conceptual issues surrounding head movement, it has been proposed that head movement is a purely postsyntactic phenomenon, i.e., as opposed to a type of syntactic movement (e.g. Chomsky, 2000, 2001; Platzack, 2013). Though the head movement process responsible for noun incorporation is excluded from this reevaluation (Chomsky, 2001, p. 37), this still raises the question of whether a head movement account of noun incorporation is necessary. Baker (2009) suggests that traditional head movement indeed co-exists with other mechanisms also responsible for incorporation cross-linguistically, and Roberts (2010) offers a novel theory of head movement (and thus incorporation) subsumed under Agree. A related possibility arising at this juncture is that the head adjunction structure in (7) may be formed without head movement. For instance, Ghomeshi and Massam (1994) and van Geenhoven (1998) posit that this configuration may be base-generated in the syntax.

Against this backdrop, Barrie and Mathieu (2012, 2016) argue that incorporation is not derived by (any theory of) head movement. They observe that incorporated nominals in many languages are often demonstrably morphologically complex, i.e., phrasal, rather than bare roots. In (9a), we see that modifier-noun sequences may incorporate in Mapudungun; (9b) additionally illustrates incorporation in Onondaga featuring a complex nominalization. According to Barrie and Mathieu, the latter cannot be derived in terms of roll-up head movement, as this would violate the Proper Head Movement Generalization of Li 1990 and Baker 1996, 2003, which bans head movement to a functional head and then a lexical head (e.g., $\sqrt{\text{-to-Voice-to-}n\text{-to-V}}$ for (9b)).³

(9) *Incorporation of phrasal XPs*

- a. adkintu-[**we-ngillan-mansun**]-kiyaw-i
care-[recently-bought-ox]-go-3.R
'S/he is caring for a recently-bought ox.' (Mapudungun; Barrie and Mathieu 2016, p. 36, citing Harmelink 1992)
- b. ho-[**at-a²ti-tshR**]-ot-a²
3SG.M.PAT-[SEMI.REFL-lean-NMLZ]-stand.upright/have-STAT
'He is using a cane.' (Onondaga; Barrie and Mathieu 2016, p. 25)

³Examples like (5), involving the incorporation of a DP into V, would likewise violate the Proper Head Movement Generalization under a head movement analysis.

[Barrie and Mathieu \(2016\)](#) propose instead that noun incorporation involves phrasal movement, as also supported by the placement of the incorporated element within the verb complex (see also [Dechaine 1999](#) for a similar idea). In the Onondaga example in (9b), the entire nominal constituent precedes the verb. If Onondaga is otherwise left-headed, as suggested by the inflectional prefixes in the language, the morpheme order of (9b) may be captured if that constituent raises to a specifier position within the vP domain. See also [Koopman and Szabolcsi 2000](#) and [Barrie 2011](#) for variations of this general analysis. Note that such a derivation would still contain instances of head movement for morpheme ordering reasons—the crucial point here is simply that noun incorporation itself is not derived by head movement.

Importantly, while this account takes phrasal movement to be a grammatically necessary step in the derivation of incorporation constructions, it is not actually the operation responsible for complex word-formation, i.e., morphological incorporation. [Barrie and Mathieu \(2016\)](#) instead suggest that certain morphologically complex (i.e., polysynthetic) languages simply prosodify clause-level syntactic structures as individual words, building on [Compton and Pittman 2010](#) (see also §4.1). Thus, the trees corresponding to (9) are ultimately spelled out as single phonological words, resulting in the appearance of incorporation. This automatically derives word-formation (including morphological incorporation) without the complex head configurations otherwise invoked in head movement accounts. See also [Ershova 2020](#) for an extension of this idea to incorporation inside complex nouns in West Circassian (Northwest Caucasian).

On the other hand, the idea that incorporation may be postsyntactic reopens the theoretical possibility of (some version of) postsyntactic head movement, especially in light of much work within Distributed Morphology unifying head movement with postsyntactic operations such as Morphological Merger ([Marantz, 1984](#); [Bobaljik, 1994](#)) or Lowering ([Embick and Noyer, 2001](#)). The account of [Harizanov and Gribanova 2019](#), for instance, takes head movement to simply be the “upward” variant of a general postsyntactic head-adjunction mechanism, which may proceed in either direction. See also [Haugen and Harley 2013](#) on interactions between head movement and other word-building operations such as Local Dislocation. As discussed in later sections,

postsyntactic word-building operations along these lines have also been utilized for both pseudo incorporation (e.g. [Levin, 2015](#); [van Urk, 2020](#)) and obligatory incorporation (e.g. [Wojdak, 2008](#); [Yuan, 2025](#)).

2.3 The status of hyponymous objects

As mentioned above, the availability of modifier stranding in incorporation constructions cross-linguistically has been interpreted as important evidence for syntactic approaches to noun incorporation. Related to this discussion is the phenomenon of *hyponymous doubling*, whereby the incorporated object co-occurs with a semantically related (usually more lexically specified) standalone object. Many (or perhaps most) languages with canonical incorporation permit both modifier stranding and hyponymous doubling, as opposed to just one or the other ([Barrie, 2015](#)).⁴ It is not immediately obvious which of the two objects is the true argument of the verb. For [Baker 1996](#), the incorporant is the true object, while the stranded element is a VP-level adjunct co-indexed with the object. As noted above, [Rosen \(1989\)](#) proposes instead that the standalone element (whether apparent modifier or hyponymous object) is the true internal argument, while the incorporant is generated inside the verb complex as a classifier.

(10) *Stranding and doubling as classifier incorporation (Kanien'kéha)*

Sh^ʔté:ku níku:ti rabahbót wahu-tsy-ahní:nu ki rake^ʔníha
 eight of.them bullhead 3M.3N-fish-bought this my.father
 ‘My father bought eight bullheads.’ ([Rosen 1989](#), p. 303, citing [Mithun 1984](#))
 Or: ‘My father fish-bought bullheads, eight of them.’

Other accounts take the two objects to be base-generated within a complex constituent. [Chung and Ladusaw \(2004\)](#) propose the structure in (11a), such that the verb’s complement is a NP that

⁴In [Baker et al. 2005](#), Southern Tiwa is cited as a language with modifier stranding and without hyponymous doubling, though the ungrammatical specific example provided therein involves two identical objects. Indeed, such constructions are ruled out by [Rosen \(1989\)](#) for independent reasons, such as the unnecessary repetition of information (though see [Haugen 2009](#) for examples of identical doubling in the Uto-Aztecan language Hopi). Inuit languages such as Kalaallisut and Inuktitut (but not their Yupik neighbors) have also been reported to lack hyponymous doubling ([Sadock, 1986, 1991](#); [Yuan, 2025](#)), though these languages have obligatory incorporation rather than canonical incorporation.

contains the incorporated nominal (NP's head) and a DP adjunct (which is ultimately stranded). As [Barrie \(2015\)](#) observes, (11a) is reminiscent of the so-called *big-DP* analysis of pronominal clitic-doubling, which likewise generates a nominal and its double (the clitic) within a complex constituent (e.g. [Torrego, 1988](#); [Uriagereka, 1995](#); [Nevins, 2011](#)). Building on Chung and Ladusaw, [Barrie \(2015\)](#) suggests that languages with hyponymous doubling either make use of the adjunction structure in (11a) or generate the nominals as the subject and predicate of a small clause, depicted in (11b).

(11) *Some possible structures for hyponymous doubling*



The analyses above generally agree that the incorporant and its hyponymous double are generated as syntactically distinct objects. Against this, however, [Haugen \(2009\)](#) proposes that these elements are actually non-identical copies of a *single* argument. Situated within the framework of Distributed Morphology ([Halle and Marantz, 1993](#)), the logic is as follows: assuming that incorporation is derived by syntactic movement and that the surface realization of movement copies arises from postsyntactic Vocabulary Insertion, some languages may permit the insertion of non-identical nominal roots into the heads and tails of movement chains. This is possible only when multiple Vocabulary Items may satisfy the featural requirements of the relevant nodes. The result of this process is the co-occurrence of two hyponymous (semantically and thematically related) nouns.

Overall, there is not a clear consensus on the grammatical relationship between the incorporated nominal and its associated standalone elements, nominal or otherwise. The existence of hyponymous doubling is thus an outstanding issue for syntactic investigations of incorporation.

3 Pseudo noun incorporation

3.1 Properties of pseudo-incorporated objects

The term ‘pseudo’ noun incorporation was coined by [Massam \(2001\)](#) to refer to constructions in which a nominal object is adjacent to a verb, though not morphologically incorporated into the verb complex. This is shown in (12), with Niuean (Austronesian). As Niuean displays base Verb-Subject-Object word order, pseudo incorporation results in Verb-Object-Subject. Additionally, the overall construction is reminiscent of canonical noun incorporation in various ways: the object is non-referential and lacks inflectional morphology (e.g., case), and the transitive subject loses its expected ERG case marking. However, pseudo incorporated objects are unambiguously phrasal (and may be syntactically quite complex), as the data in (13) illustrate.

(12) *Pseudo noun incorporation (Niuean)*

- a. Ne fakahū a tama fīfine e tohi
PST send ERG child female ABS letter
‘The girl sent the letter.’
- b. Ne fakahū tohi e tama fīfine
PST send letter ABS child female
‘The girl sent a letter.’ ([Clemens, 2019](#), p. 338)

(13) *Phrasal pseudo incorporated objects (Niuean)*

- a. Ne inu kofe kono a Mele
PST drink coffee bitter ABS Mele
‘Mary drank bitter coffee.’
- b. Ne kai sipi mo e ika mitaki a Sione
PST eat chip COM ABS fish good ABS Sione
‘Sione ate good fish and chips.’
- c. Ke kumi mena ke nonofo ai a lautolu
SBJV seek thing SBJV settle there ABS they
‘... that they would seek a place to settle.’ ([Massam, 2001](#), pp. 158-160)

There is cross-linguistic variation in the exact properties of pseudo incorporation constructions. Like Niuean, Chol (Mayan) is V-S-O and ergative (albeit in its agreement system). While pseudo incorporation also results in V-O-S, ϕ -agreement with the transitive subject still utilizes the ERG

paradigm, (14a). This contrast may be attributed to different mechanisms underlying ERG case assignment vs. ERG agreement. Languages also vary in how much inflectional morphology is still available on the object. In Hungarian (Finno-Ugric), pseudo incorporated objects may still bear overt case morphology, (14b) (e.g. [Farkas and de Swart, 2003](#)). Languages also differ regarding the accessibility of pseudo incorporated objects to ϕ -agreement—a point of variation that parallels that in canonical incorporation constructions (cf. [Baker et al. 2005](#)).

(14) *Additional pseudo incorporation patterns*

- a. Tyi i-kuch-u si' aj-Maria
PFV ERG3-carry-SS wood CFL-Maria
'Maria carried wood.' (Chol; [Coon 2010](#), p. 355)
- b. Péter **verset** olvas
Peter poem.ACC read
'Peter is reading a poem/poems/poetry.' (Hungarian; [Farkas and de Swart 2003](#), p. 96)

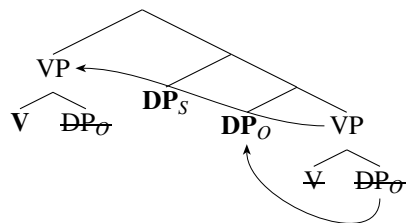
As this chapter focuses mainly on the morphosyntactic profile of pseudo incorporation constructions, I refer readers to [Farkas and de Swart 2003](#), [Chung and Ladusaw 2004](#), [Dayal 2011](#), and [Sağ 2022](#); [Sağ et al. 2025](#) for accounts of their semantic properties, as well as [Borik and Gehrke 2015](#) and [Chung and Ladusaw 2021](#) for further discussion.

3.2 Deriving verb-noun adjacency in pseudo incorporation

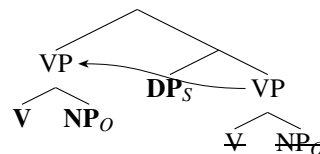
Let us now zoom in on the verb-noun adjacency illustrated in the previous examples. The syntactic research on pseudo noun incorporation has sought to account for this pattern in various ways. [Massam \(2001\)](#) proposes that pseudo incorporation in Niuean involves structurally reduced objects (e.g., NPs), which by virtue of being smaller than DPs, cannot participate in various syntactic processes (see also [Massam 2009a](#)). They are not assigned case (and could indirectly prevent ERG case assignment to the transitive subject) and cannot undergo syntactic movement. Combining this with a VP-remnant movement analysis of verb-initial word order, this derives the word order difference seen above. Specifically, DP objects undergo object shift out of the VP, which then

undergoes remnant movement, (15a); however, NP objects do not, so they raise with the VP, (15b).

(15) a. *V-S-O word order:*



b. *V-O-S word order (pseudo incorporation):*



This proposal has been applied to pseudo incorporation in various other verb-initial languages (e.g. Coon, 2010; Medeiros, 2013; Collins, 2017), though analytical details differ (see also various papers in Carnie et al. 2005). However, an important generalization about this line of analysis is that the verb-noun adjacency seen in pseudo incorporation constructions is satisfied vacuously, simply due to the NP being V's complement and being too syntactically small to move away from its in situ position. In other words, there is no sense in which the adjacency effect is derived or enforced by the grammar.

In contrast, Clemens (2014, 2019) offers an alternative account of Niuean pseudo noun incorporation that departs from the above analysis in two related ways. For Clemens, verb-initial word order is derived by head movement of the verb past the external argument, rather than object shift and VP-remnant movement. Head movement by itself yields V-S-O word order, so V-O-S word order in pseudo incorporation contexts must arise from some other mechanism. Couched within the Match Theory of Selkirk 2011, in which syntactic structures are mapped to prosodic units, Clemens proposes that pseudo incorporation in Niuean results from a prosodic well-formedness constraint, ARGUMENT- ϕ , defined in (16):

(16) *Argument Condition on Phonological Phrasing (ARGUMENT- ϕ)*

A head H with a categorial feature [C] and head C with the same [C] feature must constitute a ϕ -phrase. (Clemens 2019, p. 364)

Because head movement separates the verb and its nominal complement, ARGUMENT- ϕ is satisfied by postsyntactically reordering the object to a verb-adjacent position, as this allows the verb and object to form a single phonological phrase (ϕ). Clemens proposes that the reason why this

only affects structurally reduced objects (NPs) is because DPs are phases (Chomsky, 2001; Svenonius, 2004) and are thus spelled out separately from the verb; assuming that the DP's category feature is rendered invisible by spell-out, ARGUMENT- ϕ is vacuously satisfied. See also Clemens and Coon 2018 and van Urk 2025 for cross-linguistic extensions of this analysis.

Both of the above accounts are concerned with the adjacency between the verb and its NP complement—but are agnostic about the position of the *head* of the NP relative to the verb. Other cross-linguistic investigations, however, have revealed that verb-noun adjacency pertains truly to *head-head adjacency*. In the Niuean examples in (13) above, the nominal heads of the pseudo incorporated complex NPs happen to be immediately adjacent to the preceding verb, since the modifiers within these NPs are postnominal. However, there are verb-initial languages that permit both prenominal and postnominal modifiers, such as Tongan (Austronesian). Importantly, only postnominal modifiers are permitted when the nominal is pseudo incorporated, (17b).

(17) *No modifier between verb and pseudo incorporated noun (Tongan)*

- a. *Na'e **tō** ki'i **manioke** 'a Sione
PST plant small cassava ABS Sione
'Sione planted a small amount of cassava.'
- b. Na'e **tō** **manioke** iiki 'a Sione
PST plant cassava small ABS Sione
'Sione planted a small amount of cassava.' (Ball, 2005, pp. 12-13)

Such patterns may also be seen in non-verb-initial languages, as discussed by Baker (2014). Tamil (Dravidian) displays base Subject-Object-Verb order, so whether an object is pseudo incorporated is not immediately obvious on the basis of word order alone. Baker (2014) proposes that, in contrast to ACC-marked objects, as in (18a), bare objects in languages like Tamil are pseudo incorporated: they are interpreted as non-specific indefinites and may not be separated from the verb by intervening adverbs, (18b).

(18) *Case and word order alternations (Tamil)*

- a. Maala **anda pustagatt-e** veegamaa **paḍi-cc-aa**
Mala the book-ACC quickly read-PST-3FS
'Mala read the book quickly.'

- b. Maala {veegamaa} **pustagam** {*veegamaa} **paḍi-cc-aa**
 Mala quickly book quickly read-PST-3FS
 ‘Mala read a book/books quickly.’ (Baker, 2014, p. 8)

The verb-noun adjacency requirement shown in (17) above is also evident in Tamil. Resultatives in Tamil typically follow the object on which they are predicated, but this becomes impossible on unmarked, i.e., pseudo incorporated, objects, (19). Baker further demonstrates that the ill-formedness of (19) may be rescued by displacing the PP elsewhere in the syntax.

(19) *Resultative and direct object word order (Tamil)*

- *Baala **pustagam** meese kiiḷe **va-kkir-aan**
 Bala book table under put-PRES-3MS
 Intended: ‘Bala puts book(s) under the table.’ (Baker, 2014, p. 11, 13)

To handle these facts, Baker (2014) adopts a head movement analysis, building on his prior work (e.g. Baker, 1988; Baker et al., 2005). The basic idea here is that syntactic head movement has consequences at both LF and PF. Head movement creates a complex predicate that is interpreted as a single semantic predicate in the sense of Dayal 2011; it also ensures head-head adjacency if the verb-adjacent copy of the head movement chain is pronounced. See also Levin 2015 for further developments of this idea, to be further discussed below.

That being said, it is worth noting that some of the empirical generalizations for least Tamil have been disputed. Driemel (2020) reports that certain focus adverbs and particles may intervene between pseudo incorporated nouns and verbs, as in (20). Driemel thus suggests that not all case-unmarked objects are genuinely pseudo incorporated, a conclusion echoed by Barrie and Kang (2022) and Barrie (2024) for Mongolian (though these authors nonetheless acknowledge that such bare objects are still syntactically more restricted than ACC-marked ones).

(20) *Certain intervening adverbs permitted between bare object and verb (Tamil)*

- Maala **pustagam**-kuuṭa **paḍi-cc-aa**
 Mala book-MIR read-PST-3FS
 ‘Mala even read a book/books.’ (Driemel, 2020, p. 7)

A question that arises here is *why* verb-noun adjacency is required in pseudo incorporation constructions. For [Baker \(2014\)](#), head movement is optional; the aforementioned interpretive and ordering effects affecting the noun arise only when it does take place. A different approach, however, holds that verb-noun adjacency *is* grammatically enforced under certain conditions—and that it serves a purpose complementary to the traditional Case Filter ([Levin, 2015](#), et seq.).

The Case Filter was originally motivated by the observation that nominal elements may be distributionally restricted, in that there are certain syntactic positions in which they may not appear (e.g. [Vergnaud, 1977](#); [Chomsky, 1981, 2000](#)). Nominal arguments are therefore systematically ruled out in positions where they are inaccessible to a local case-assigning head (e.g., finite T, transitive *v*). Against this backdrop, a number of authors have shown that certain patterns resembling pseudo noun incorporation likewise display constraints on the distribution of nominals, suggesting that pseudo incorporation may license nominals when case assignment is not possible.

Consider the Balinese (Austronesian) data below, based on the discussion of [Levin 2015](#). As in many other Austronesian languages, Balinese displays a voice system that tracks the syntactically most prominent argument. In the object voice, the subject remains *in situ* in a postverbal position. Like Tongan and Tamil, Balinese permits various clausal and noun-internal word orders—but constructions consisting of an *in situ* subject and a selecting verb are systematically ruled out whenever the two are non-adjacent, (21).

(21) *Verb-noun adjacency with in situ subjects (Balinese)*

- a. {sanget-sanget} gamelan-e {sanget-sanget} **depak** { *sanget-sanget } **ia**
 very-very gamelan-DEF very-very OV.hit very-very 3
 {sanget-sanget}
 very-very
 ‘S/he was hitting the gamelan really hard.’ ([Levin, 2015](#), p. 94)
- b. Nyoman **gugut** { *liu } **cicing** {liu}
 Nyoman OV.bite many dog many
 ‘Many dogs bit Nyoman.’ ([Erlewine, Levin, and van Urk, 2017](#), p. 393)

Balinese thus also displays a verb-noun adjacency requirement, though the nominals involved are not pseudo incorporated in the typical sense, since they are neither direct objects nor necessarily

non-specific indefinites (as in (21a)).⁵ [Levin \(2015\)](#) nonetheless proposes a common analysis for Balinese *in situ* subjects and pseudo incorporation cross-linguistically: both have incomplete extended projections, lacking the outermost K(ase)P layer implicated in traditional case assignment ([Bittner and Hale, 1996](#)). Verb-noun adjunction (whether via head movement or a postsyntactic operation such as Merger or Lowering) allows KP-less nouns to become part of the extended projection of the verb, thus obviating the Case Filter. In Balinese, only *in situ* subjects are impacted due to the unavailability of structural NOM case assignment in object voice contexts ([Erlewine, Levin, and van Urk, 2017](#)). This approach is thus similar in spirit to [Baker’s \(1988\)](#) analysis of (canonical) noun incorporation as an alternative to case-licensing, though note that the very idea that a postsyntactic word-formation mechanism may obviate the Case Filter suggests that the Case Filter itself is not necessarily imposed in the syntax, contrary to the traditional theory.

This proposal has been extended to various other verb-noun adjacency patterns seen cross-linguistically (e.g. [Erlewine, Levin, and van Urk, 2017](#); [van Urk, 2020](#); [Fong, 2024](#)), including ones that do not involve pseudo incorporation to begin with; [Branan \(2022\)](#), for instance, captures word order restrictions in Kikuyu (Bantu) using the same licensing-by-adjacency logic as above. On the other hand, see [Driemel and Tebay 2022](#) for an alternative analysis of Balinese that does not invoke licensing (these authors instead attribute the above patterns to feature-driven structure removal in the sense of [Müller 2017, 2018](#)). If non-licensing accounts are viable for other attested verb-noun adjacency effects, then the relationship between (pseudo) incorporation and nominal licensing remains an open question.

4 Obligatory incorporation

4.1 Incorporation via affixal verbs

‘Obligatory’ incorporation is also referred to in the literature as constructions containing lexical affixes (e.g. [Mithun, 1997](#)), denominal verbs (e.g. [Gerdtz and Hukari, 2008](#); [Stonham, 2008](#)), or

⁵That being said, see [Öztürk 2009](#) and [Sağ et al. 2025](#) on agent pseudo incorporation in Turkish.

affixal predicates (e.g. [Wojdak, 2008](#); [Muro, 2009](#)), all indicating that noun incorporation is triggered by a morphological condition on the verbs. In such languages, there is a closed class of verbs that *require* a morphologically incorporated object, as in (22), from Labrador Inuttut (Inuit-Yupik-Unangan). The verbs in question are thus characterizable as affixes, in that they may not surface without a morphological host. In the Inuit languages, these verbs co-exist with other (non-affixal) verbs that never trigger incorporation. However, in other languages such as Hopi ([Haugen, 2009](#)) and Ojibwe and Onondaga ([Barrie and Mathieu, 2016](#)), obligatory and canonical (optional) incorporation co-exist.

(22) *Obligatory incorporation (Labrador Inuttut)*

- a. **pitsi-tu-vunga**
dried.fish-consume-IND.1SS
'I'm eating dried fish.'
- b. ***pitsi-mik** **tu-vunga**
dried.fish-MODALIS consume-IND.1SS
Intended: 'I am eating dried fish.' ([Johns, 2007](#), p. 541)

As noted in §2.1, it has been debated whether obligatory incorporation is a subtype of incorporation to begin with. [Mithun \(1986\)](#) posited that obligatory incorporation is outside of the scope of the typology of noun incorporation of [Mithun 1984](#), given the morphologically bound nature of the verbal elements (see also [Baker 1988](#)). However, the literature on incorporation now does generally take obligatory incorporation to be a subtype of noun incorporation, given that the incorporants often exhibit the same morphosyntactic and semantic properties as in canonical incorporation constructions (e.g. [Sadock, 1980, 1986](#); [Bok-Bennema and Groos, 1988](#); [Johns, 2007, 2009](#); [Wojdak, 2008](#); [Wiltschko, 2009](#); [Yuan, 2025](#)).

The majority of the work on obligatory incorporation has focused on Inuit languages such as Kalaallisut and Inuktitut, and is in broad agreement that the incorporated nominals are true thematic objects of the incorporating verbs. As first demonstrated by [Sadock \(1980\)](#), these nouns are referential, that is, able to antecede pronouns introduced later in the discourse, (23a) (see also [Johns 2007, 2009](#); [Yuan 2025](#)). They also underlyingly encode morphosyntactic case and number,

as indicated by the concord morphology on stranded modifiers, (23b), even though case and plural morphology is not encoded on the incorporants themselves.

(23) *Obligatorily incorporated nouns are syntactic objects (Kalaallisut)*

- a. Suulut **biili**-lior-sima-voq. Sikaat karsi-kua-n-nik
Søren.ABS car-make-PERF-IND.3SS. cigar.GEN.PL box-old-3P-MODALIS
sana-sima-**vaa**
make-PERF-IND.3SS/3SO
‘Søren has made a car_i. He made it_i with old cigar boxes.’
- b. ataatsi-**nik** **qamuti**-qar-poq
one-MODALIS.PL carriage-have-IND.3SS
‘He has one carriage [*plurale tantum* in Kalaallisut].’ (Sadock, 1980, pp. 309, 311)

Sadock (1985, 1991) thus argues that obligatory incorporation involves both syntax and morphology, and develops his Autolexical Theory largely on this basis. In this theory, the syntactic and morphological grammatical modules operate simultaneously but independently, resulting in non-isomorphisms between their respective representations. The syntax creates true nominal objects of verbs, but the morphology dictates that these objects surface as incorporated due to the affixal nature of the verbs selecting them. This explains why the incorporants still display the same structural and interpretive properties otherwise associated with standalone arguments.

Other analyses posit instead that obligatorily incorporated nouns in Inuit (and elsewhere) are syntactically smaller than their standalone counterparts (e.g. Johns, 2007; Compton and Pittman, 2010), in accordance with their morphologically bound and generally uninflected appearance. Such approaches assume a more direct correspondence between syntax and morphology, and take incorporants to be akin to their canonically and pseudo incorporated counterparts in other languages (as discussed above). Johns (2007) proposes that obligatory incorporation in Inuktitut arises due to a syntactic requirement targeting roots. Johns argues that affixal verbs are light verbs (of category *v*) and are inherently rootless; incorporation of nominal roots into affixal verbs is needed to fulfill this requirement. In contrast, non-incorporating lexical verbs do contain roots so they never trigger incorporation. An interesting piece of evidence for this analysis comes from the existence of so-called ‘dummy roots’, akin to the English expletive ‘it’, syntactically inserted when no other root

is found, (24).

(24) *Dummy roots in obligatory incorporation (Inuktitut)*

- a. **pi-qa-nngit-tuq**
DUMMY-have-NEGATION-DECL.3SS
'He has nothing.'
- b. **pi-gasuaq-tuq**
DUMMY-make.an.effort-DECL.3SS
'He is trying.' (Johns, 2007, p. 559-560)

Johns (2007) ultimately concludes that the (non-)occurrence of incorporation ultimately boils down to the syntactic distinction between functional vs. lexical items in the verbal domain (see also Cook and Johns 2009). Broadly, affixal verbs in languages with obligatory incorporation indeed appear to be semantically more general than non-affixal verbs, lending support for this idea (e.g. Mithun 1999, Barrie and Mathieu 2016, though see Huijting 2024 for some counterexamples).

A potential complication for Johns' account, however, is that the nominal element selected by the affixal verb may be morphologically complex—potentially much larger than just a bare root, as in (25). For Johns (2007, fn. 27), the additional suffixes inside the incorporant must be analyzed as functional elements, even when they have apparently derivational meanings. Additionally, affixal verbs may also serve as restructuring predicates, i.e., incorporate complex verbal constituents, (26) (e.g. Pittman, 2006, 2009; Yuan, 2026). Note that (26b) illustrates an affixal verb attaching to another affixal verb (which, in turn, attaches to a nominal argument).

(25) *Complex obligatorily incorporated noun (Inuktitut)*

- a. **[uqa-limaar-vi]-liu(ng)-inna-nngit-tunga**
[speak/tongue-all.of-NMLZ.LOC]-make-always-NEG-DECL.1SS
'I was not always making libraries.' (Compton and Pittman, 2010, p. 2173)
- b. **[iglu-tsiava-nngua]-qaq-tuq**
[house-great-pretend]-have-DECL.3SS
'S/he has a great pretend house.' (Compton, 2013, p. 3)

- (26) *Complex obligatorily incorporated non-nominal constituents (Inuktitut)*
- a. Miali **igla-qu-jara**
 Mary.ABS laugh-order-DECL.1SS/3SO
 ‘I ordered Mary to laugh.’ (Pittman, 2009, p. 1)
- b. Jamesie-up **[[tuktu-mini]-tu]**-nira-qqau-janga Jaani
 Jamesie-ERG [[caribou-former]-consume]-say-REC.PST-DECL.3SS/3SO Jaani.ABS
 ‘Jamesie said that Jaani ate caribou meat.’ (Yuan 2026, p. 9)

As first mentioned in §2.2, [Compton and Pittman \(2010\)](#) provide a different analysis for the Inuit languages, in which the functional vs. lexical distinction is orthogonal to incorporation. Subsumed under their broader account of complex word-formation, this approach generalizes that standalone words are largely syntactically either CPs (verbs, which in Inuit necessarily bears clause type morphology) or DPs (case-marked nominal arguments), and theorizes that syntactic phases in these languages are prosodified as individual phonological words upon spell-out. If incorporated nominals are always structurally reduced, i.e., smaller than DPs, then they are accordingly not phasal and cannot be pronounced on their own. Likewise, the restructuring predicates in (26) are analyzed as embedding vP- or TP-sized complements ([Pittman, 2006, 2009](#)). For [Compton and Pittman \(2010\)](#), then, obligatory incorporation results from the size of the object, not the affixal or functional status of the verb.⁶ This account is thus compatible with canonical incorporation too and has been adopted for such patterns accordingly (e.g. [Barrie and Mathieu, 2016](#); [Ershova, 2020](#)).

Conversely, Yuan (2025) concurs with Sadock 1985, 1991 that incorporated nouns in the Inuit languages are not syntactically reduced but rather structurally equivalent to standalone objects. Yuan does not adopt Sadock’s Autolexical Theory, proposing instead that a Distributed Morphology model, in which the syntax serves as the input to the morphological component, is equally compatible with this idea. If the postsyntactic operation responsible for complex word-formation may apply to full (non-reduced) objects generated by the syntax, then we would get morphologi-

⁶Note that this analysis, if taken literally, does not immediately account for the possibility of incorporated nouns of category D, such as pronouns, wh-elements, and proper names. However, we can readily take the relevant nominal phase to be the highest projection of the extended nominal domain, i.e., K(ase)P rather than DP. The incorporation of CP-sized clauses has also been argued to be attested in the Inuit languages (Yuan, 2024, 2026); for such constructions to be accommodated under Compton and Pittman’s (2010) account, we might likewise such clauses to be missing the clausal superstructure associated with full (phasal) clauses (e.g., ForceP in the terminology of Rizzi 1997).

What remains unexplained under Yuan's account is why incorporated nouns are morphologically uninflected if they are syntactically full DPs. Nonetheless, evidence for this theory comes from the crucial observation that incorporated nouns, at least in Inuktitut, are syntactically active. The examples below show that these nominals may be targeted by ϕ -morphology and co-occur with ERG-marked subjects, (27a), and may additionally undergo phrasal movement (see also [Johns 2009](#)). These facts stand in contrast to the typologically more familiar picture of syntactically inert incorporants. The latter is illustrated in (27b-c) with passivization, which may create new antecedents for binding, and relativization, which allows incorporants to take scope outside of the relative clauses containing them.

a. Taiviti-up **sivalaar-tu-ruma-jangit**

b. **aasivar**-tuq-ta-u-juq nuliaqta-**mi**-nut
spider-consume-PASS.PART-be-DECL.3SS mate-REFL.POSS-ALLAT
'The spider_i is being eaten by its_i mate.'

Yuan's (2025) account additionally makes a novel typological prediction: other languages with obligatory incorporation should also have syntactically active incorporants, if incorporation in these cases is also triggered by morphological conditions that have nothing to do with the syntactic behavior of the nouns themselves. This is borne out in Nuu-chah-nulth (Wakashan), which also has affixal verbs and permits its incorporated objects to undergo passivization and relativization, (28) (Wojdak, 2005, 2008):

(28) *Phrasal movement of obligatorily incorporated nouns (Nuu-chah-nulth)*

- a. **k^waq-**'iic-ck^wi-'at-ʔiiš ʔuuš-qh-'at
 spawned.herring.eggs-consume-EVID-PASS-3.IND some-do.by-PASS
 'Spawned herring eggs must have been eaten by someone.'
- b. k^winʔaʔ-mit-ʔiiš John ʔučʔin [**yaq-**'aap-mit-ʔiitk]
 like-PST-3.IND John dress REL-buy-PST-2SG.RL
 'John liked the dress you bought.' (Wojdak, 2008, pp. 57, 201)

There is an interesting point of variation between the two languages concerning the behavior of modifiers of incorporated nouns. In Inuktitut, these modifiers are necessarily stranded,⁷ (29a), but, in Nuu-chah-nulth, they must be incorporated, resulting in the nominal arguments being stranded instead, (29b). Wojdak (2008) posits that the latter results from the predicate-initial (Verb-Object-Subject) word order of the language, coupled with a postsyntactic incorporation mechanism sensitive to linear adjacency ("PF incorporation"). The element that immediately follows the verb complex incorporates, whether that be the nominal object or a prenominal adjective modifying the object.

(29) *Modifier vs. nominal stranding in obligatory incorporation*

- a. **guulu-mik** **ujami**-taa-ruma-junga
 gold-MODALIS necklace-get-want-DECL.1SS
 'I want to get a gold necklace.' (Inuktitut; Yuan 2025, p. 551)
 (*ujaming-mik guulu-taa-ruma-junga)
- b. **haʔum-**'iic-ʔiiš-ʔaʔ **ʔaapinis**
 tasty-consume-3.IND-PL apples
 'They are eating delicious apples.' (Nuu-chah-nulth; Wojdak 2008, p. 44)
 (*ʔaapinis-'iic-ʔiiš-ʔaʔ haʔum)

Yuan suggests that this difference may boil down to the relative timing of postsyntactic word-formation in the individual languages (e.g. Embick and Noyer, 2001). If word-formation takes place relatively early in the postsyntax in Inuktitut, this explains its sensitivity to structural notions such as syntactic complementation and ability to skip over modifiers (adjuncts). However, if word-formation occurs late enough to lose access to syntactic information, as Wojdak (2008) assumes

⁷That being said, the incorporation of modifiers is optionally permitted in the Nunavik varieties of Inuktitut (Beach, 2011). See Yuan 2025, pp. 553-554 for discussion.

for Nuu-chah-nulth, we correctly expect it to only operate on linear adjacency.

4.2 Obligatory incorporation due to other factors

Obligatory incorporation may also be triggered by morphosyntactic properties unrelated to those of the selecting verb. Wiltschko (2009) analyzes a class of lexical suffixes in Halkomelem (Salish) as incorporated acategorical roots in the sense of Marantz 1997; these elements are obligatorily incorporated in that they are morphologically bound. Wiltschko takes them to be modifiers that may double standalone arguments, (30), and proposes that they must attach to the verb due to a root-root compounding process.

- (30) *Root incorporation (Halkomelem)*
 th'éxw-wíl-t-es te ló:thel
 wash-dish-TRANS-3S DET dish
 'He washed the dish.' (Wiltschko, 2009, p. 211)

In other languages, incorporation is required for nominals with particular morphosyntactic features. In Fijian (Oceanic), a predicate-initial language, proper names and pronouns in object position must undergo pseudo incorporation, meaning that they are necessarily verb-adjacent and appear without a preceding article, (31) (e.g. Aranovich, 2013; van Urk, 2020). This is typologically unusual in that the pseudo incorporated objects are higher-ranked in the prototypical animacy scale (Silverstein, 1976; Aissen, 2003) and might be expected to be *more* morphologically marked than lower-ranked objects.⁸ Similar feature-specific obligatory incorporation patterns are found in related languages, such as Iai (Pearce, 2001) and Tongan (Otsuka, 2000).

- (31) *Obligatory incorporation of high-ranked objects (Fijian)*
 e a kau-ti Jone / au mai ko Eroni
 3SG PST bring-TR.PR Jone 1SG DIR ART.PR Eroni
 'Eroni brought Jone/me.'

On this basis, Aranovich (2013) characterizes Fijian as a 'partial' polysynthetic language un-

⁸The pseudo incorporation pattern shown here for Fijian co-exists with a canonical incorporation process that affects other objects, but the two constructions are morphosyntactically distinguishable.

der the definition of polysynthesis of [Baker 1996](#). According to this idea, all standalone (non-incorporated) nouns in polysynthetic languages are actually adjuncts, the true argument positions licensed by either verbal agreement or noun incorporation (cf. [Jelinek, 1984](#)). Aranovich proposes that, while most nouns in Fijian are syntactically dislocated, proper names and pronouns are true verbal complements, explaining why only they appear verb-adjacent. Fijian is therefore partially polysynthetic because some nouns are adjuncts while others are not.

However, [van Urk \(2020\)](#) demonstrates that proper names and pronouns in syntactically derived positions also obey this adjacency effect, even though they are thematically unrelated to the preceding verb and may be separated by a clause boundary, (32). The observation that non-arguments of a verb may still require adjacency casts doubt on the partial polysynthesis characterization above. Van Urk instead proposes that pseudo incorporation in Fijian reflects licensing-by-adjacency in the sense of [Levin 2015](#) (see §3.3), required for arguments higher on the animacy scale as a Differential Object Marking strategy. Note also that the data in (32), which show $\bar{A}$ -movement to the embedded clause edge, further reinforce [Yuan's \(2025\)](#) observation that obligatorily incorporated nouns may often be syntactically active.

(32) *Pseudo incorporation of nominals in derived positions (Fijian)*

- a. au **kila-a** [CP **Eroni** e na sure-ti Jone]
 1SG know-TR.N Eroni 3SG FUT invite-TR.PR Jone
 'I know Eroni will invite Jone.'
- b. au a gai **kila-i** [CP **cei** e talei-taka ko Eroni] nanoa
 1SG PST GAI know-TR.PR who 3SG like-TR.N ART.PR Eroni yesterday
 'I found out yesterday who Eroni likes.' ([van Urk, 2020](#), pp. 338-339)

Finally, Southern Tiwa (Tanoan) is another language with obligatory incorporation triggered by the properties of the nominals involved ([Allen et al., 1984](#); [Rosen, 1990](#); [Heck and Richards, 2010](#)). However, the empirical picture is highly complex, and depends not only on the properties of the incorporants themselves but also on the nature of any structurally higher arguments present. It is therefore not obvious that Southern Tiwa may analyzed along the same lines as the other languages discussed above. In intransitive clauses, inanimates must incorporate, while animates must not,

(33a-b), whereas objects of transitive clauses must incorporate (regardless of animacy), (33c), barring a systematic set of exceptions. One such exception pertains to the morphosyntactic features of the transitive subject: object incorporation becomes optional rather than obligatory if the subject is 1st or 2nd person, (33d).⁹

(33) *Obligatory and non-obligatory incorporation (Southern Tiwa)*

- | | |
|--|--|
| <p>a. We-fan-lur-mi
C.NEG-snow-fall-PRES.NEG
'Snow is not falling.'
(*Fan we lur-mi)</p> | <p>c. *Seuanide Ø-mũ-ban
man A:A-see-PST
Intended: 'S/he saw the man.'
(instead: Ø-seuan-mũ-ban)</p> |
| <p>b. Musade we-seur-mi
cat C.NEG-fall-PRES.NEG
'The cat is not falling.'
(*We-musa-seur-mi)</p> | <p>d. Seuanide ti-mũ-ban
man 1SG:A-see-PST
'I saw the man.'
(also ok: Ti-seuan-mũ-ban)
(Rosen, 1990, pp. 680-681, 683-684)</p> |

Rosen (1990) accounts for these facts using a fixed feature ranking that contains grammatical functions and person features, coupled with an autosegmental approach that links the individually ranked elements with association lines rooted in a different tier. Under this approach, all ungrammatical combinations in Southern Tiwa, including (33c), result from crossing association lines. However, incorporated arguments are immune because they are simply not targeted by association lines, so incorporation may circumvent otherwise ill-formed structures.

Heck and Richards (2010), however, note that this ban on crossing lines is reminiscent of *intervention effects* arising from standard locality conditions on Agree (Chomsky, 2000, 2001), in that a probe cannot skip a potential goal in order to target a less local one. These authors also observe that the person asymmetry in (33c-d) resembles the Weak Person-Case Constraint (PCC) effect (cf. Bonet, 1991). While the PCC is more typically seen with combinations of pronominal clitics, what these have in common is that the choice of direct object is restricted when a higher argument (e.g., a subject) is 3rd person but not when it is 1st or 2nd person. Putting these ideas together, Heck and

⁹Restrictions on (non-)incorporation are also found in ditransitive clauses (though the exact permissible combinations differ). Incorporation also interacts with the agreement morphology on the verb. Here, I focus just on incorporation in transitives for reasons of space, but see the citations above for descriptions and analyses of the ditransitive patternings and the agreement restrictions.

Richards reanalyze the Southern Tiwa data in terms of restrictions on the accessibility of DPs to higher Agreeing probes, building on existing intervention-based analyses of the PCC that follow the same logic (e.g. [Anagnostopoulou, 2003, 2005](#); [Rezac, 2004](#)). Incorporated nouns do not face this issue because they are structurally reduced NPs that lack the outermost DP layer.

Both [Rosen \(1990\)](#) and [Heck and Richards \(2010\)](#) are primarily focused on explaining the restrictions on person combinations, and thus mainly view incorporation in transitive contexts such as (33c) as a ‘repair’ for such restrictions: incorporation circumvents an otherwise ungrammatical structure. Under this treatment, the obligatoriness of incorporation is not a grammatical requirement *per se*: standalone objects may be generated, as in (33d), so long as their presence does not crash the derivation. This, however, does not explain the obligatory incorporation of inanimates in intransitive clauses, as in (33a); for both analyses, the latter requirement is an arbitrary fact about the language. A question for future research, then, is whether there is an underlying principle in Southern Tiwa that unites incorporation in both intransitive and transitive contexts.

Overall, obligatory incorporation remains relatively understudied as a cross-linguistic phenomenon, compared to its canonical and pseudo incorporation counterparts. Although it has received attention in the literature on Inuit and a handful of other languages, more work is needed to determine whether the grammatical characteristics identified above are representative of obligatory incorporation more generally. Similarly, it is not yet clear what broad generalizations we may form about the triggers for obligatory incorporation or the properties of the incorporants themselves.

5 Conclusion

This chapter has overviewed the major patterns of incorporation found across the world’s languages, with particular emphasis on canonical, pseudo, and obligatory noun incorporation. From an empirical standpoint, the study of incorporation has primarily been propelled by investigations into lesser-studied (e.g., non-European) languages, and has considerably widened our knowledge of syntactic typology. On the theoretical side, as noted by [Massam \(2009b, p. 1051\)](#), incorpora-

tion “provide[s] us with a microcosm of linguistic theory”, in that it raises important and difficult questions about the relationship between syntax, morphology, and the lexicon.

The empirical and analytical findings summarized above show that it is far from clear whether incorporation is a singular, uniform phenomenon; perhaps we should instead view it as a family of related constructions whose properties continue to refine our understanding of grammar. As such, incorporation remains a productive testing ground for theories of syntax and its interfaces and showcases the importance of diverse cross-linguistic evidence in the development of general linguistic theory.

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